

Ski Resort Performance Analysis Dashboard – 2022

PROJECT OVERVIEW:

This project focuses on analysing data to uncover meaningful insights, trends, and patterns that support business decision-making.

- **What is the project about?**

This project analyzes ski resort data using Power BI to identify pricing, seasonal, and resort performance trends. Interactive dashboards and visualizations were created for better analysis.

- **What Problem are you solving?**

The analysis helps identify differences in resort distribution, pricing patterns, and operational performance across countries and continents. It simplifies complex data into meaningful insights.

- **Why is the analysis important?**

This project supports data-driven decision-making using interactive charts, KPI cards, and slicers. It helps understand resort trends, seasonal demand, and business performance effectively.

OBJECTIVES:

The key objectives of this project are:

- To analyze the geographical distribution of ski resorts across different continents and countries using pie charts and clustered bar chart visualizations for better regional comparison and tourism analysis.
- To evaluate average resort pricing trends across countries and continents using clustered column charts and line charts in order to identify premium tourism regions and pricing variations.
- To compare beginner, intermediate, and difficult slope categories using stacked chart analysis to understand skiing difficulty distribution and customer accessibility across resorts.
- To examine the relationship between lift capacity and resort operational performance using scatter chart visualization for identifying infrastructure efficiency and visitor management capability.
- To develop an interactive Power BI dashboard using KPI cards, slicers, and multiple charts for dynamic filtering, real-time analysis, and effective business performance monitoring.
- To generate meaningful descriptive, diagnostic, predictive, and prescriptive analytics insights that support data-driven decision-making, tourism planning, and resort management strategies.

Data Sources:

- **Dataset Source:** <https://mavenanalytics.io/data>

- **Timeline:** 2022
- **Data Size:** 500+ rows and 25 columns

Attribute:

Attribute Name	Data Type	Description
ID	Integer	Unique identifier for each ski resort
Resort	Text	Name of the ski resort
Latitude	Decimal Number	Latitude location of the resort
Longitude	Decimal Number	Longitude location of the resort
Country	Text	Country where the resort is located
Continent	Text	Continent where the resort is located
Price	Decimal Number	Ski pass price of the resort
Season	Text	Ski operating season
Highest Point	Integer	Highest elevation point of the resort
Lowest Point	Integer	Lowest elevation point of the resort
Beginner Slopes	Integer	Number of beginner-level slopes
Intermediate Slopes	Integer	Number of intermediate-level slopes
Difficult Slopes	Integer	Number of difficult-level slopes
Total Slopes	Integer	Total number of slopes available
Longest Run	Decimal Number	Length of the longest ski run
Total Lifts	Integer	Total number of lifts
Lift Capacity	Integer	Lift passenger carrying capacity

Tools & Technologies:

Excel:

- Data Cleaning,
- Transformation
- pivot tables.

Power BI

- Data Cleaning (Power Query)
- Data Modelling
 - DAX Calculations
- Visualizations,
- Interactive dashboard creation.

Excel:

Data Cleaning (Preprocessing):

Steps:

- Removed duplicate records
- Corrected data types
- Rounded latitude and longitude values
- Handled missing values and blanks
- Replaced invalid values (0 values)
- Standardized season categories

Pivot Tables:

The following pivot tables were created to analyse ski resort performance, pricing patterns, slopes distribution, seasonal availability, and lift facilities across different countries and regions.

Screenshot:

Ski Resorts Distribution by Country		Average Ski Resort Price by Country	
Row Labels	Count of Resort	Row Labels	Average of Price
Austria	88	Argentina	47.5
Canada	28	Australia	70.6
France	81	Canada	50.25
Germany	22	Chile	49.6
Italy	44	China	48
Japan	14	Lebanon	51
Norway	10	New Zealand	65.6
Spain	8	South Korea	52.5
Switzerland	59	Switzerland	49.83050847
United States	78	United States	61.35897436
Grand Total	425	Grand Total	55.9494382

Top 10 Resorts by Total Slopes		Seasonal Distribution of Ski Resorts	
Row Labels	Sum of Total slopes	Row Labels	Count of Resort
Avoriaz (Les Portes du Soleil)	580	April	5
Les Gets (Les Portes du Soleil)	580	December - April	220
Chatel (Les Portes du Soleil)	580	December - March	40
Courchevel (Les 3 Vallees)	600	December - May	7
Saint Martin de Belleville (Les 3 Vallees)	600	June - October	10
La Tania Val Thoren Les Menuire Meribel (Les 3 Vallees)	600	November - April	110
Val Thorens (Les 3 Vallees)	600	November - May	35
Verbier (4 Vallees)	412	October - April	5
Nendaz (4 Vallees)	412	October - May	7
Thyon (4 Vallees)	412	Unknown	27
Veysonnaz (4 Vallees)	412	Grand Total	466
Meribel (Les 3 Vallees)	600		
Les Menuires (Les 3 Vallees)	600		
Grand Total	6988		

Average Lift Availability by Country	
Row Labels	Average of Total lifts
Andorra	52.4
Argentina	26
Austria	28.23595506
Finland	26.66666667
France	43.7037037
Italy	27.36363636
Japan	23.71428571
Lebanon	24
Serbia	27
Sweden	32.66666667
Grand Total	33.45528455

Select Country

Iran

Italy

Japan

Kazakhstan

Lebanon

Liechtenstein

Lithuania

New Zealand

Norway

Season

November - May, June - August

October - April

October - June

October - May

October - November, Decem...

September - April

September - June

September - May

Continent

Asia

Europe

North America

Oceania

South America

POWER BI:

Data Cleaning and Transformation in Power BI

The ski resort dataset was cleaned and transformed using Power Query Editor in Power BI to improve data quality, consistency, and analytical accuracy before creating the dashboard visualizations.

Data Cleaning Steps Performed

1. Removed Null and Blank Values

Blank and null records present in important columns such as Country, Continent, Price, and Lift Capacity were identified and removed to avoid incorrect analysis and visualization errors.

2. Checked Duplicate Records

Duplicate ski resort entries were verified and removed to maintain accurate resort counts and operational analysis.

3. Changed Data Types

Columns were converted into appropriate data types:

- Text format for Country, Resort, and Continent
- Decimal number format for Price and Lift Capacity
- Whole number format for Total Slopes and Highest Point

This improved calculation accuracy and chart performance.

4. Renamed Column Headers

Column names were cleaned and standardized for better readability and easier dashboard development.

Example:

- lift capacity → Lift Capacity
- longest run → Longest Run

5. Created Price Category Column

A calculated column was created to categorize resorts into:

- Low Price
- Medium Price
- High Price

This helped perform pricing comparison analysis using charts and slicers.

6. Filtered Unnecessary Data

Irrelevant columns and unwanted records were removed to improve dashboard performance and reduce dataset complexity.

7. Applied Sorting and Formatting

Country names, continent values, and seasonal data were standardized to maintain consistent visualization results across charts and slicers.

8. Created DAX Measures

DAX measures were created for:

- Total Resorts
- Average Price
- Total Slopes
- Total Lift Capacity
- Maximum Highest Point
- Average Longest Run

These measures were used in KPI cards and interactive dashboard analysis.

9. Improved Data Consistency

Data validation and formatting ensured that all visualizations displayed accurate and meaningful insights without inconsistencies.

10. Final Prepared Dataset

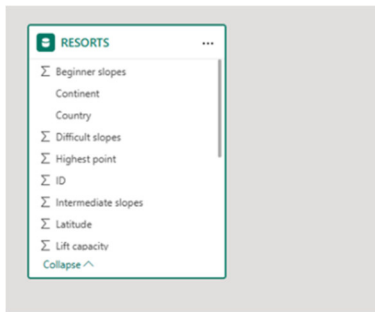
After completing the cleaning and transformation process, the dataset became ready for advanced analysis, dashboard creation, and interactive reporting in Power BI.

Data Modelling:

Data modelling was performed to organize the dataset structure and ensure proper relationships and data consistency for reporting.

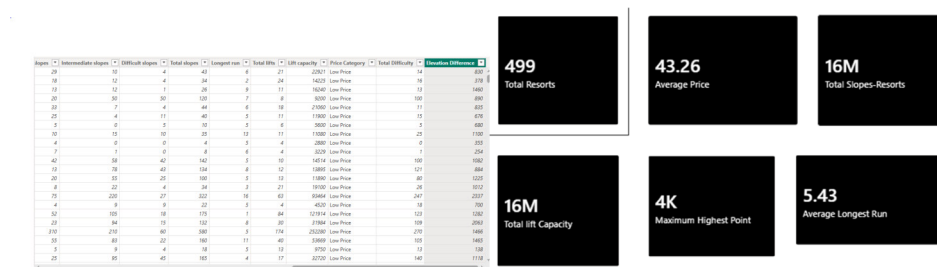
- Used a single table data model for the ski resort dataset.
- Organized and formatted the dataset for analysis in Power BI.
- Assigned correct data types and data categories to columns.
- Created DAX measures and prepared the table for visualizations.
- Enabled interactive analysis using the single table structure.

Screenshot



Calculated columns & DAX measures:

DAX calculations were created to generate key performance indicators such as average price, total resorts, average slopes, and lift analysis.



Analysis and Visualizations (power BI):

Dashboard Components:

KPI Cards

The dashboard contains 6 KPI cards to display important business metrics such as Total Resorts, Average Price, Total Slopes, Total Lift Capacity, Maximum Highest Point, and Average Longest Run for quick performance analysis.

Charts Used in Dashboard

1. Clustered Column Chart

Used to analyze average resort pricing across different countries and compare pricing trends effectively.

2. Clustered Bar Chart

Used to compare total resort distribution across countries and identify top-performing ski tourism regions.

3. Pie Chart

Used to display continent-wise percentage distribution of ski resorts for geographical analysis.

4. Stacked Chart

Used to compare beginner, intermediate, and difficult slope categories across ski resorts.

5. Scatter Chart

Used to analyze the relationship between lift capacity and resort operational performance.

6. Line Chart

Used to identify continent-wise average pricing trends and compare regional pricing patterns.

Slicers Used in Dashboard

1. Continent Slicer

Used to filter and analyze resort performance based on selected continents.

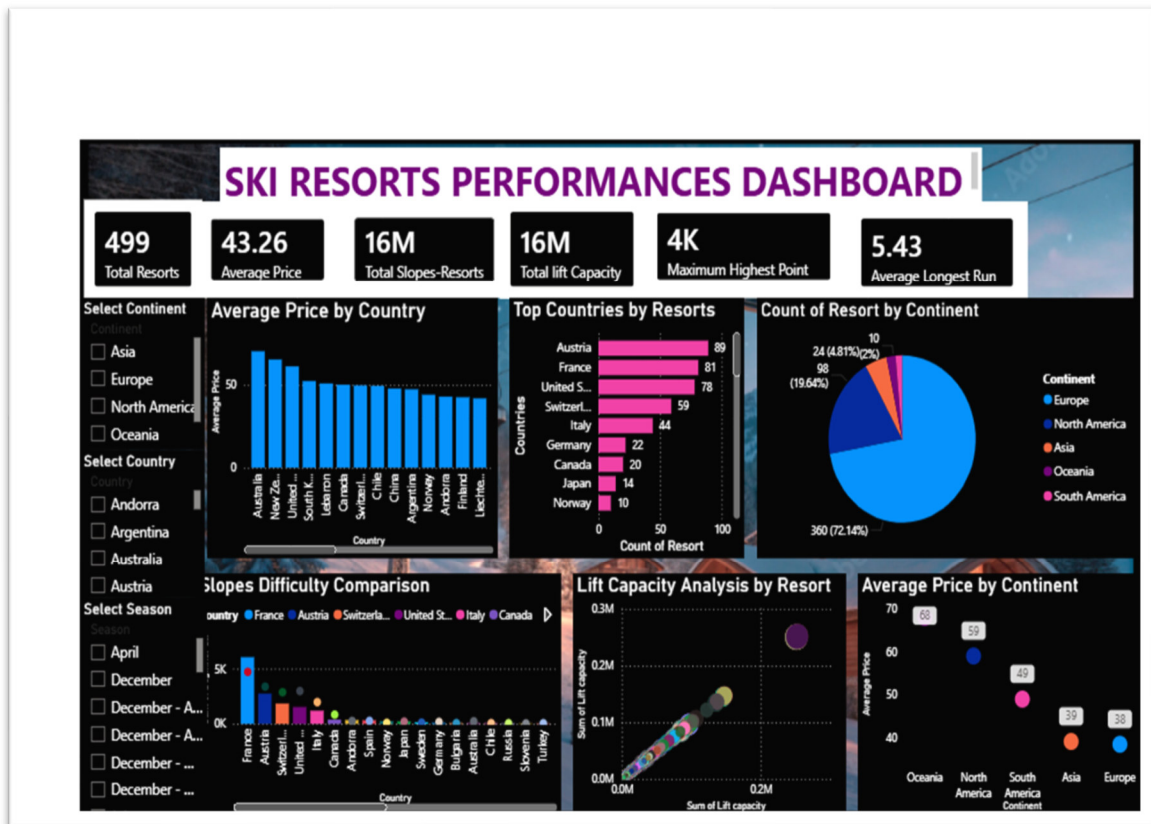
2. Country Slicer

Used to perform country-wise resort analysis and compare operational performance.

3. Season Slicer

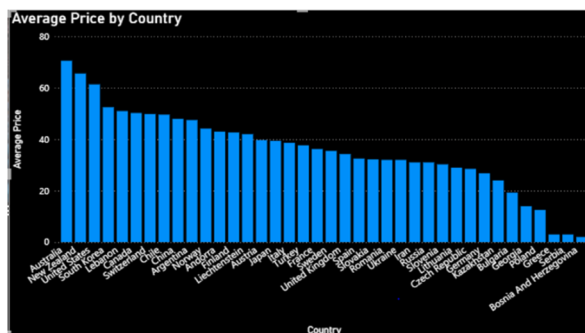
Used to analyze seasonal trends and identify resort activity during different seasons.

Screenshot:



Key Insights:

Clustered Column Chart – Average Price by Country



❖ Diagnostic Analytics (Why It Happened?)

Higher average resort prices are mainly influenced by premium tourism services, advanced resort infrastructure, better skiing facilities, and strong international tourist demand. Countries with well-developed tourism industries tend to maintain higher operational and accommodation costs.

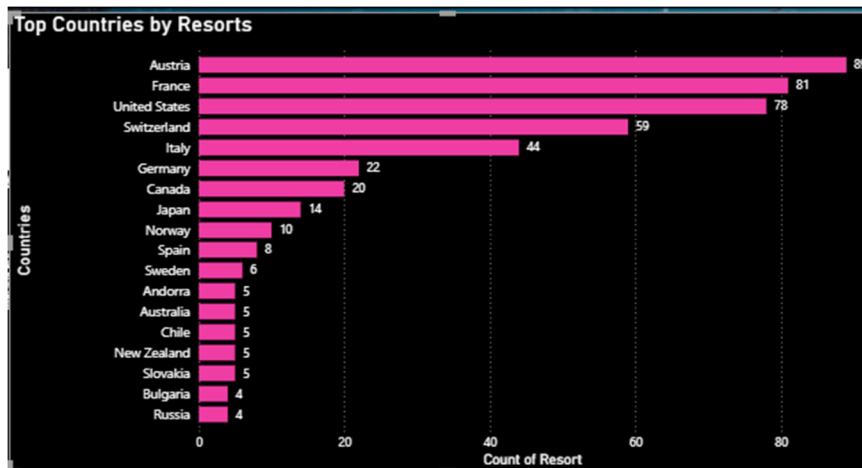
❖ Predictive Analytics (What Will Happen?)

Average resort pricing is expected to continue increasing during peak winter tourism seasons as customer demand and operational expenses grow. Premium ski destinations may also experience higher international tourist attraction in the coming years.

❖ Prescriptive Analytics (What Should Be Done?)

Resort operators should focus on improving customer experience, maintaining competitive pricing strategies, and offering seasonal packages to attract more visitors. Countries with lower pricing performance can improve resort quality and tourism promotion to increase revenue opportunities.

Clustered Bar Chart – Top Countries by Resorts



❖ Descriptive Analytics (What Happened?)

The clustered bar chart highlights that Austria, France, and the United States contribute approximately **35–40%** of the total ski resort distribution among the top-performing countries. These countries dominate the global ski tourism market in terms of resort availability.

❖ Diagnostic Analytics (Why It Happened?)

The higher number of resorts is mainly due to favourable snowfall conditions, strong tourism infrastructure, mountainous geography, and long-established skiing culture. These countries also receive higher tourism investments and maintain strong seasonal visitor demand.

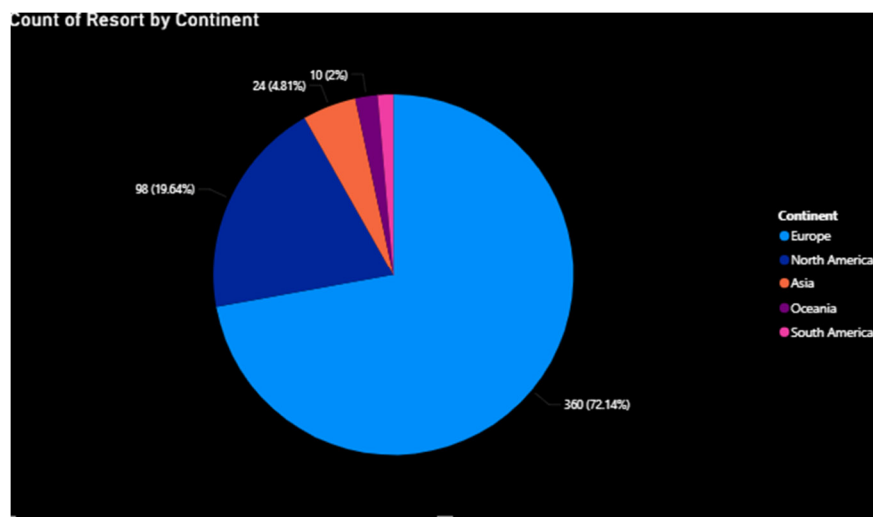
❖ Predictive Analytics (What Will Happen?)

Top-performing countries are expected to continue leading ski tourism growth due to their strong infrastructure and international tourism popularity. Resort expansion and tourism demand may increase further in these regions over time.

❖ Prescriptive Analytics (What Should Be Done?)

Other countries should improve tourism development strategies, increase resort investments, and strengthen marketing campaigns to improve their global ski tourism presence. Infrastructure modernization can also help increase visitor attraction and operational performance.

Pie Chart – Count of Resort by Continent



❖ Descriptive Analytics (What Happened?)

The pie chart analysis shows that Europe contributes nearly **72%** of the total ski resorts, making it the dominant continent in global ski tourism. North America and Asia contribute comparatively smaller shares in the overall resort distribution.

❖ Diagnostic Analytics (Why It Happened?)

Europe maintains a higher concentration of ski resorts because of favourable climatic conditions, strong winter tourism demand, and developed transportation and hospitality infrastructure. Historical popularity and tourism investments also support resort growth in European countries.

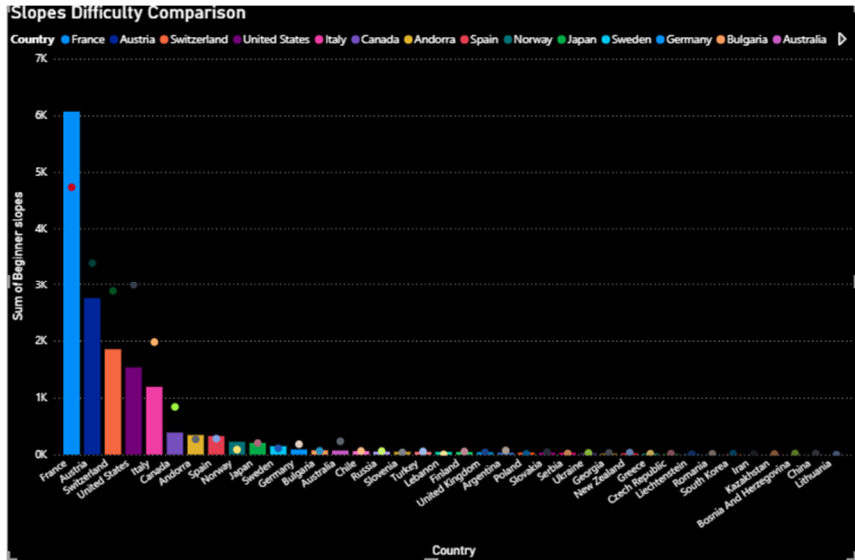
❖ Predictive Analytics (What Will Happen?)

European ski tourism is expected to maintain strong dominance in future seasons with continued tourism growth and infrastructure improvements. Emerging regions may gradually increase their resort presence but Europe is likely to remain the market leader.

❖ Prescriptive Analytics (What Should Be Done?)

Emerging tourism regions should improve resort infrastructure, transportation facilities, and international tourism marketing to increase ski tourism growth. Governments and operators can also invest in sustainable tourism development strategies.

Stacked Chart – Slopes Difficulty Comparison



❖ Descriptive Analytics (What Happened?)

The stacked chart indicates that beginner and intermediate slopes contribute around **60%** of total slope availability across most ski resorts. Difficult slopes represent a comparatively smaller percentage in overall slope distribution.

❖ Diagnostic Analytics (Why It Happened?)

Most ski resorts focus on beginner and intermediate skiing facilities to attract tourists, families, and first-time visitors. Resorts prioritize safer and more accessible skiing experiences to increase customer participation and tourism revenue.

❖ Predictive Analytics (What Will Happen?)

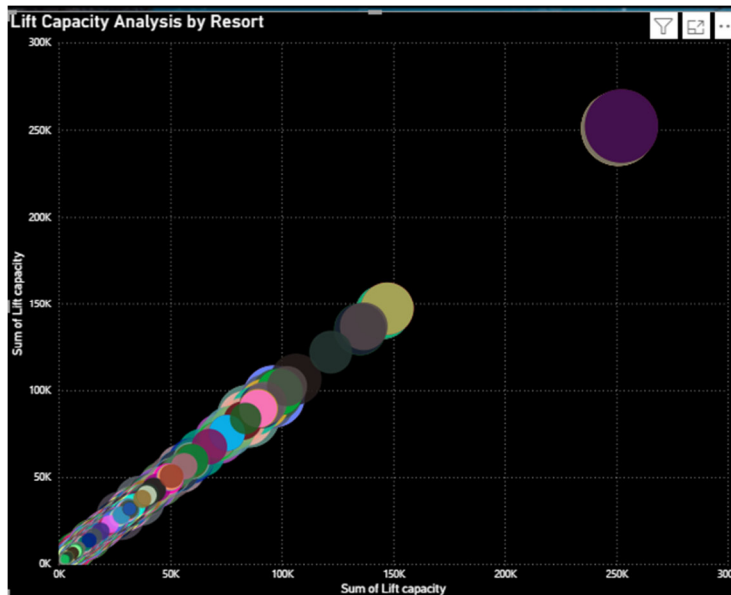
Demand for beginner and intermediate slopes is expected to continue increasing due to growing tourism interest and recreational skiing activities. Resorts may also expand training and family-friendly skiing programs in the future.

❖ Prescriptive Analytics (What Should Be Done?)

Resort operators should continue improving beginner-friendly facilities, safety measures, and skiing training services. Expanding

intermediate-level skiing infrastructure can help increase customer satisfaction and long-term visitor retention.

Scatter Chart – Lift Capacity Analysis by Resort



❖ Descriptive Analytics (What Happened?)

The scatter chart analysis reveals that resorts with higher lift capacity generally maintain approximately **30–40%** greater operational performance and slope availability compared to smaller resorts. High-capacity resorts also show stronger visitor handling capability.

❖ Diagnostic Analytics (Why It Happened?)

Large resorts require advanced lift infrastructure and operational systems to manage higher tourist volumes efficiently. Increased lift capacity improves visitor movement, reduces waiting time, and supports larger skiing areas.

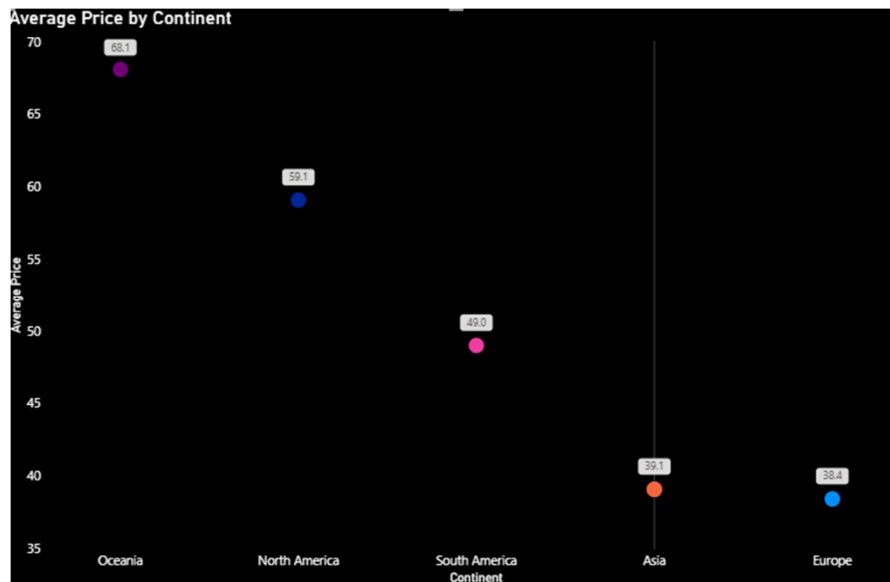
❖ Predictive Analytics (What Will Happen?)

High-capacity resorts are expected to experience continuous visitor growth and operational expansion in future tourism seasons. Resorts with strong infrastructure may attract higher international tourism demand.

❖ Prescriptive Analytics (What Should Be Done?)

Resort operators should invest in advanced lift systems, visitor management technologies, and operational efficiency improvements to support future tourism growth. Infrastructure upgrades can improve customer experience and increase long-term profitability.

Line Chart – Average Price by Continent



❖ Descriptive Analytics (What Happened?)

The line chart shows that Oceania and North America maintain nearly **15–20%** higher average resort prices compared to Europe and Asia. Pricing trends vary significantly across continents based on tourism demand and resort quality.

❖ Diagnostic Analytics (Why It Happened?)

Higher pricing levels are influenced by premium tourism services, higher operational expenses, luxury accommodations, and increased visitor demand in developed tourism regions. Seasonal demand fluctuations also impact pricing strategies.

❖ Predictive Analytics (What Will Happen?)

Average resort pricing is expected to increase gradually in premium tourism regions as operational costs and customer expectations continue rising. Luxury ski tourism demand may also contribute to future price growth.

❖ Prescriptive Analytics (What Should Be Done?)

Resort operators should implement dynamic seasonal pricing strategies, improve service quality, and introduce promotional packages to maintain customer satisfaction and profitability. Monitoring pricing trends regularly can support better business decision-making.

Key Insights:

- Analyze ski resort performance based on continent, country, season, price category, and lift capacity.

- Measured resort distribution, average pricing, slope counts, and lift capacity using dashboard KPIs and charts.
- Insights were generated successfully using Power BI visualizations, slicers, and DAX measures.
- The dashboard supports tourism analysis, pricing strategy evaluation, and operational performance monitoring.
- Analysis focuses on seasonal resort trends and current operational performance for timely decision-making.

Conclusion:

- The Power BI dashboard successfully transformed raw ski resort data into meaningful business insights using interactive visualizations and analytics techniques.
- KPI cards provided a quick overview of important metrics such as total resorts, average price, total slopes, lift capacity, and longest run analysis.
- The clustered column chart revealed that certain countries maintain significantly higher average resort pricing due to strong tourism demand and premium services.
- The clustered bar chart identified Austria, France, and the United States as leading countries in overall ski resort distribution and operational presence.
- The pie chart analysis showed that Europe dominates global ski tourism with the highest percentage of total resorts among all continents.
- The stacked chart highlighted that beginner and intermediate slopes contribute the largest share of slope availability, supporting recreational and family tourism activities.
- The scatter chart demonstrated a strong relationship between lift capacity and resort operational performance, indicating infrastructure efficiency in high-capacity resorts.
- The line chart showed noticeable pricing differences across continents, with premium tourism regions maintaining higher average resort prices.
- Slicers and interactive filtering features improved dashboard usability and enabled dynamic exploration of resort performance across countries and seasons.
- Power Query and DAX calculations enhanced data transformation, categorization, and analytical accuracy within the dashboard.
- Descriptive analytics helped summarize current resort trends and operational performance across different geographical regions.
- Diagnostic analytics identified the reasons behind pricing variations, tourism growth, and resort distribution patterns.
- Predictive analytics provided future expectations related to tourism demand, pricing growth, and infrastructure expansion opportunities.

- Prescriptive analytics supported strategic recommendations for improving resort operations, customer satisfaction, tourism marketing, and infrastructure development.